

AWM-Fuse: Multi-Modality Image Fusion for Adverse Weather via Global and Local Text Perception

Xilai Li¹, Huichun Liu, Xiaosong Li², Tao Ye³, Zhenyu Kuang⁴, and Huafeng Li⁵

Abstract—Multi-modality image fusion (MMIF) in adverse weather aims to address the loss of visual information caused by weather-related degradations, providing clearer scene representations. Although a few studies have attempted to incorporate textual information to improve semantic perception, they often lack effective categorization and thorough analysis of textual content. To address these limitations, we propose AWM-Fuse, a unified fusion framework that handles diverse weather degradations via global and local text perception with shared parameters. In particular, a global text perception module leverages BLIP-generated captions to extract overall scene features and identify primary degradation types, thus promoting generalization across various adverse weather conditions. Complementing this, the local module employs detailed scene descriptions produced by ChatGPT to concentrate on specific degradation effects through concrete textual cues, enabling the recovery of subtle details. Furthermore, textual descriptions are used to constrain the generation of fused images, effectively steering the network learning process toward better alignment with semantic labels, thereby promoting the learning of more meaningful visual features. To facilitate text-guided fusion under adverse weather, we construct AWM-Text, a large-scale benchmark providing paired global and local annotations for multi-modality image pairs. Extensive experiments demonstrate that AWM-Fuse consistently outperforms state-of-the-art methods under complex weather conditions and on multiple downstream tasks. Our code is available at <https://github.com/Feecuini/AWM-Fuse>

Index Terms—Multi-modality image fusion, adverse weather, text-guided fusion, unified framework.

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I. INTRODUCTION

INFRARED and visible image fusion (IVIF) aims to generate more expressive image representations by utilizing information from multiple sources [1], [2], [3], [4], [5]. Such fused representations benefit downstream tasks, including object detection [6] and semantic segmentation [7]. Under adverse weather (e.g., rain, haze, and snow), visible images often exhibit severe texture degradation and semantic ambiguity caused by scattering, occlusion, and reduced contrast [8]. In contrast, infrared images are less sensitive to changes in illumination and can provide relatively stable target contours and thermal information. However, infrared alone is insufficient for fine-grained semantic understanding because it lacks color information and rich high-frequency textures. Therefore, it is crucial to recover weather-degraded details while enabling effective cross-modal interaction. A unified adverse-weather fusion framework can exploit modality complementarity to reduce the uncertainty introduced by single-modality degradation, yielding more robust scene semantics and improving downstream performance.

However, image fusion under adverse weather conditions remains a significant challenge. The difficulty lies not only in the deterioration of image quality, but also in the complex and inconsistent nature of the degradation, which significantly disrupts the correspondence between infrared and visible images. For example, rain and snow usually appear as sparse local streaks or occlusions, while haze causes global contrast drop and color fading [9]. In some cases, infrared images also lose contrast. As a result, the feature distributions of the two modalities change, and local structures become less consistent. Cross-modal alignment and interaction then become more sensitive to noise. This often produces mismatched content or artifacts in the fused image. Therefore, it is hard to train a single shared-weight model that works well across different weather conditions while still recovering useful details. This motivates the use of higher-level semantic guidance. Under extreme degradation conditions, relying solely on low-level pixel constraints often enables the model only to restore local textures or overall brightness, but is unable to reliably distinguish useful semantic details from weather degradation, making it difficult to stably generate semantically consistent fusion results. Based on this observation, introducing text as explicit semantic cues can provide clearer content constraints and strategic guidance for the fusion process, helping the

model more robustly restore key semantic information under complex degradation.

Recent advances in Vision-Language Models (VLMs) have demonstrated the effectiveness of using textual semantics to guide visual modeling [2], [10], [11], [12], [13], [14], [15]. Text provides a structured semantic framework by offering explicit descriptions that highlight key image content. Incorporating textual descriptions into visual modeling helps the model link objects to relevant attributes (e.g., their states and context), improving its ability to capture and interpret image content [16]. This strategy has also been introduced to the image fusion task [2], [10], where text-guided methods have shown improved multi-modality feature interaction and facilitated user interaction. Despite efforts to integrate VLMs into fusion and restoration for better real-world performance, several challenges remain:

(1) Limited design of textual embeddings: Existing studies [2], [17] often use simple textual cues to represent degradations, without fully leveraging the expressive outputs of VLMs. In practice, this approach treats discrete labels or short prompts (e.g., “rainy” or “hazy”) as a single global vector, which is then used to modulate fusion weights or feature channels. However, complex scenes exhibit both global degradation and nuanced local variations in object color, motion, and blur, which are often missed by simplified descriptions.

(2) Overemphasis on local details: Subsequent works [10], [18] generated detailed scene descriptions from multiple image labels and used them as stronger conditioning cues to enable the model to restore details according to the descriptions. However, this approach often leads to excessive focus on fine details, while neglecting global contrast restoration or overall semantic consistency. When text is incomplete or the scene is unseen, the model tends to generalize poorly, reducing robustness in real-world fusion.

(3) Limited robustness under adverse weather: Even recent VLM-based fusion models [2], [10] struggle under adverse weather, where severe conditions can introduce multiple degradations in visible images and may also reduce contrast in infrared images. Moreover, it remains challenging to maintain performance across diverse weather patterns using a single shared-weight model.

In summary, existing VLM-based text-guided fusion methods still face challenges under adverse weather, as they often fail to exploit the hierarchical structure of text. Moreover, joint degradations in visible and infrared images exacerbate cross-modal uncertainty, making it difficult to produce fusion results that are both semantically consistent and robust with a single shared-weight model. To address these limitations, we propose AWM-Fuse, a multi-modality image fusion method designed for adverse weather. We leverage text at both global and local levels and exploit its semantic content more effectively to guide fusion under adverse weather. Specifically, the global text perception module uses BLIP [19] to generate a concise caption that summarizes the scene structure, main targets, and the overall degradation type. This caption provides stable global semantics, helping the model decide what to emphasize or suppress when the multi-modal input is severely degraded.

Meanwhile, the degradation caused by rain and snow typically manifests as local and shape-varying obstructions, which can mask high-frequency textures and introduce local artifacts. Since these changes often occur only in certain areas of the image and have strong randomness, a global caption alone may fail to cover these local details, making it prone to missing local changes that are crucial to fusion quality. Therefore, the local text perception module uses ChatGPT [20] to generate detailed descriptions of target attributes, motion states, and local degradations. These descriptions complement the global caption and guide more precise cross-modal interaction and detail recovery in salient regions. Through the complementary design of “global constraint + local supplement”, the model can produce high-quality fused results when both global and local degradation coexist, thereby avoiding the problems in existing VLM fusion methods such as “insufficient description leading to information deficiency” or “excessive details leading to overfitting”. For global captions, we use the CLIP text encoder to extract text features [21], which keeps the caption features in the same embedding space as our CLIP-based semantic supervision. However, CLIP text embeddings mainly capture sentence-level global semantics and may be less sensitive to fine-grained local degradation cues in long descriptions. Therefore, for the detailed text generated by ChatGPT, we use the BLIP text encoder, which better preserves token-level and sequential information for fine-grained semantic alignment.

Existing fusion methods [4], [22], [23], [24], [25] often prioritize visual modality and lack semantic alignment constraints. In contrast, CLIP, a powerful VLM pre-trained on large-scale image-text data, effectively captures image-text alignment. Leveraging this capability, we propose a novel VLM-driven loss function that utilizes the image-text matching of CLIP to align the fusion output with the textual description of a clean multi-modality image during training. This image-text feature matching learning strategy enhances the robustness and adaptability of fused images across diverse weather conditions and degradation scenarios, improving real-world applicability. We also construct AWMM-Text, a large-scale text benchmark with paired textual annotations for adverse-weather multi-modality image pairs. Our contributions are summarized as follows:

- We propose AWM-Fuse, a novel multi-modality image fusion method for adverse weather. AWM-Fuse leverages global and local text perception within a unified, shared-weight architecture to handle diverse degradations.
- We propose novel global and local text perception modules that guide feature extraction from both macro and micro perspectives, enhancing generalizability while preserving high-fidelity fusion.
- We propose a VLM-driven loss to enforce image-text alignment, and we construct AWMM-Text, the first benchmark with paired global and local annotations for adverse-weather multi-modality images.
- We conduct extensive experiments under diverse adverse weather conditions, showcasing the state-of-the-art performance of AWM-Fuse and its practical applicability.

in real-world scenarios through various downstream tasks.

II. RELATED WORK

A. Multi-Modality Image Fusion

In recent years, researchers [4], [26], [27], [28], [29], [30], [31], [32] have focused on harnessing the synergy between infrared and visible modalities in IVIF. However, because infrared and visible images come from different modalities, simple feature fusion can lead to pixel redundancy or over-reliance on one modality, diverging from human visual perception. To address this, decoupling shared and modality-specific features has become a widely adopted strategy. For example, CDDFuse [4] combines the global attention of Transformer with the unique local context extraction capabilities of CNN. Recently, researchers [24] have also explored higher-order modeling for IVIF tasks or have revisited the fusion generation process from an equal-variance perspective [3]. Overall, these methods provide effective solutions for encouraging richer and more efficient multimodal interaction, enabling the fused representation to better exploit complementary information across modalities.

To enhance the real-world applicability of IVIF, many studies have adapted it for downstream tasks [23], [33], [34], [35], [36]. For example, Liu et al. [23] developed a two-layer optimization approach to jointly address fusion and target detection, enabling the preservation of salient target information in fusion results. Tang et al. [35] introduced a semantically-aware real-time fusion network, combining fusion with semantic segmentation to enrich fused images with semantic details. Additionally, deploying IVIF algorithms [37], [38], [39], [40], [41], [42] in complex scenes has garnered significant attention. For instance, Li et al. [8] explored a unified framework that combines the image restoration and fusion tasks, and introduced the first IVIF architecture specifically tailored for adverse weather conditions. To cope with the effects of real-world noise, Huang et al. [41] proposed an image decomposition-driven network to jointly perform fusion and denoising tasks. Li et al. [9] proposed a unified fusion model based on light transmission decomposition. It estimates light scattering in the visible image to separate degradation from useful information, then extracts salient cues and reconstructs the fused result. Jie et al. [42] proposed a joint fusion and super-resolution method with semantic guidance and clarity awareness. It treats both tasks as conditional generation and uses clarity-aware semantics for adaptive low-resolution handling and cross-modal feature extraction. Xu et al. [43] proposed an unsupervised degradation-robust fusion network with intrinsic content extraction, content fusion, and appearance transfer. It fuses degradation-free content features and applies high-quality appearance statistics to generate the final result. Except for AWFusion [8], most methods do not study IVIF under adverse-weather degradations, although they are common in real scenes. AWFusion unifies weights via distillation but does not promote a consistent representation across degradation types and levels. In contrast, we first introduce

VLM-based semantic guidance for adverse-weather IVIF to unify fusion and degradation handling, improving generalization.

B. Vision-Language Model for Low-Level Vision Tasks

With the rapid advancement of deep learning, VLMs are becoming increasingly relevant to low-level vision tasks, showcasing strong performance across various applications [10], [11], [12], [13], [18], [44], [45]. Representative VLMs such as CLIP, BLIP, and ChatGPT-4 [20] offer complementary strengths: CLIP aligns visual and textual representations, enabling cross-modal retrieval and semantic interpretation of images; BLIP generates descriptive captions and performs visual question answering, thereby enhancing scene-level understanding. ChatGPT-4 strengthens contextual reasoning and natural language generation. Together, these models highlight the potential of VLMs to bridge vision and language, expanding the possibilities for vision tasks guided by linguistic cues. For example, Zhao et al. [10] proposed a VLM-based image fusion framework that guides the fusion process using text information in the source image. Luo et al. [12] introduced a multi-tasking framework for image restoration using a degradation-aware VLM. Zhang et al. [46] proposed a language-driven semantic robust image fusion method. This method utilizes the location verification information provided by VLM and dynamically aggregates multi-modal features under the guidance of language instructions, thereby achieving preference modulation and highlighting of specific targets. Li et al. [47] proposed a language-driven multi-degradation perception image fusion model that introduces semantic priors to adaptively modulate multimodal features in the channel dimension, thereby achieving adaptive enhancement strategies for different degradation types. Cao et al. [48] proposed an efficient image fusion algorithm based on language and mask guidance, mainly using the generated caption encoding as text features, and alternately concatenating them with the image feature sequence at different layers of the network, thereby encouraging the model to retain key information related to semantics during fusion. Although the above methods have demonstrated the effectiveness of language priors in fusion tasks, they usually treat the text as a single-grained semantic condition, lacking hierarchical modeling of global and local semantic cues. In contrast, the proposed method classifies and utilizes semantic cues from both the global and local text levels, achieving more robust and semantically consistent high-fidelity fusion in adverse weather conditions.

III. METHODOLOGY

A. Overview

We propose a unified IVIF framework for all-weather adverse scenarios, centred on multi-scale textual modelling. Under degraded conditions such as rain, haze, or snow, pure visual features often prove unstable, prone to target loss or semantic misalignment. Fortunately, textual information offers more robust high-level semantic cues that serve as effective

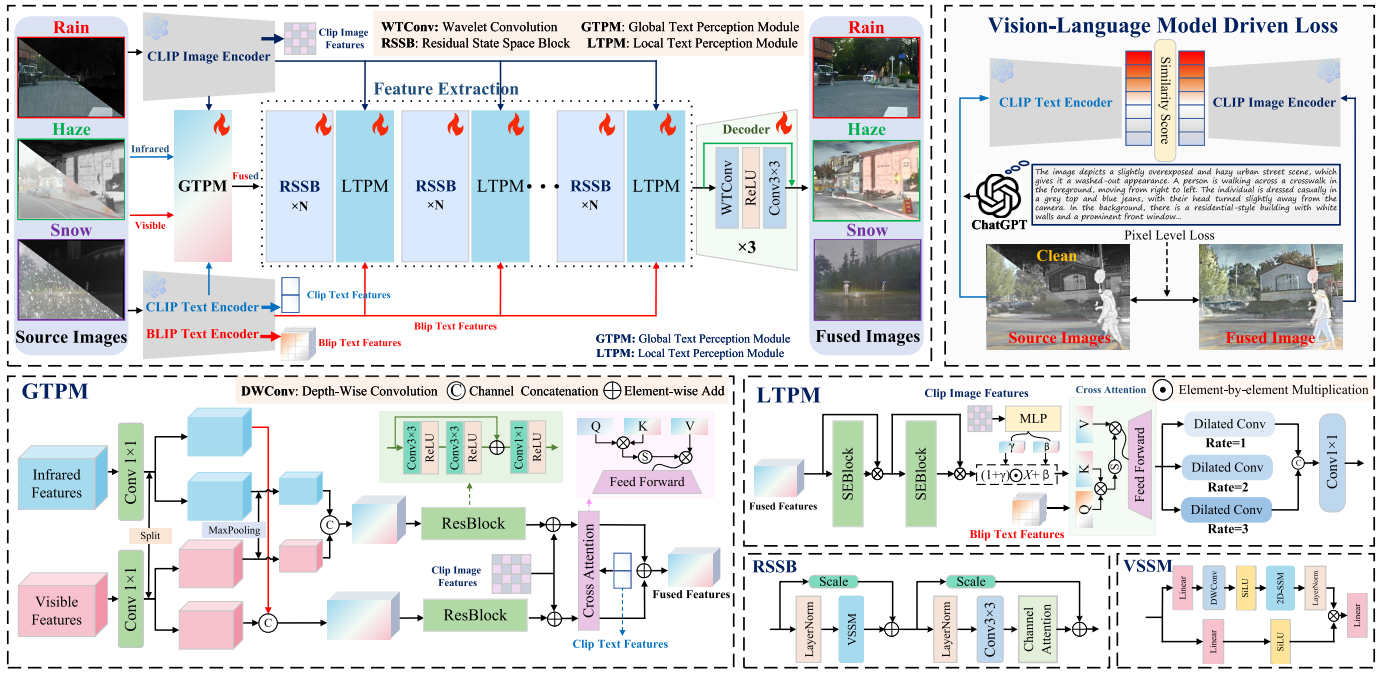


Fig. 1. The architecture of the proposed multi-modality image fusion method for adverse weather.

priors to augment visual data. Accordingly, we inject complementary textual semantics into the visual fusion process: (i) *Global Text Perception* models holistic scene semantics, providing scene-level priors to enhance model robustness under severe degradation; (ii) *Local Text Perception* focuses on fine-grained entities, attributes, and their local relationships, guiding detail fidelity and structural reconstruction; (iii) *Semantic Consistency Supervision* constrains the fused visual representations to align with textual semantics, thereby suppressing potential semantic drift during fusion. The overall workflow is illustrated in Figure 1.

Our method consists of four main components: (1) Global Text Perception Module (**GTPM**), (2) Residual State Space Block (**RSSB**) [49], (3) Local Text Perception Module (**LTPM**), and (4) the decoder based on Wavelet Convolution (**WTConv**) [50]. The GTPM is designed to establish a deep correlation between image features and global textual features, enhancing the ability of our model to perceive and adapt to various degraded scenes. Mamba provides a potential solution for balancing the global receptive field with computational efficiency, enabling the modeling of long-range dependencies with minimal computational cost [51], [52]. We use the RSSB as an intermediate feature extraction module, effectively addressing spatial information loss and channel redundancy issues introduced by the flattening strategy in the traditional Mamba module [49]. The LTPM, positioned directly after the RSSB, enhances the contextual understanding of the model by guiding it with detailed semantic information. This approach ensures semantic consistency in the fused features while improving the extraction of finer details.

The image decoder is expected to efficiently reconstruct holistic image content. However, small kernel convolutions (e.g., 3×3) limit the receptive field of CNN. Inspired by large-kernel convolutional networks [53], [54], [55], we find that

increasing the kernel size is more effective than stacking additional layers. Thus, we incorporate WTConv in the decoder. By extracting features in the wavelet domain, WTConv effectively enlarges the receptive field without significantly increasing the parameter count.

B. Global Text Perception Module

To enhance the ability of the proposed algorithm to handle various degraded scenes, we employ the GTPM, which takes BLIP-generated captions as text input and integrates features encoded from the multi-modality source image via CLIP to supplement scene features. The main process of GTPM is illustrated in Figure 1. First, a 1×1 convolution expands the feature channels of the source image to obtain the visible feature Img_{vi} and the infrared feature Img_{ir} . This expansion not only increases the representational capacity of the feature space but also facilitates subsequent feature interaction. Next, the infrared and visible feature maps are split into two parts for multi-scale fusion.

$$Img_m^L, Img_m^R = \text{Split}(Img_m) \quad (1)$$

where Img_m represents the image features with $m = vi$ for visible and $m = ir$ for infrared, L and R denote the left and right halves respectively. For the left half, infrared and visible features are concatenated directly to produce the first set of fused features $Fuse^L$, while the right half undergoes max pooling before concatenation to obtain the second set $Fuse^R$, thereby capturing hierarchical and multi-scale contextual information. This decomposition allows the model to simultaneously leverage fine-grained details and broader contextual dependencies, enhancing the robustness of fusion across diverse degradation patterns. This operation can be formally expressed as follows:

$$Fuse^L = \text{Concat}(Img_{vi}^L, Img_{ir}^L) \quad (2)$$

$$Fuse^R = \text{MAP}(\text{Concat}(Img_{vi}^R, Img_{ir}^R)) \quad (3)$$

where $\text{MAP}(\cdot)$ denotes the max-pooling function. These two sets of fused features are then processed by a residual block to enhance feature representation and minimize feature loss. The residual block effectively mitigates potential gradient vanishing issues during fusion and improves feature robustness. The fused features are subsequently integrated with image features extracted by the CLIP encoder, ensuring consistency across multi-modality information. To incorporate semantic information and improve model comprehension, we employ a cross-attention module [10] to fuse text and image features. In this setup, text features serve as queries Q_{Text} , while image features act as keys K_{Fuse} and values V_{Fuse} , enabling the model to adaptively locate the relevant image regions corresponding to each text feature. It can be expressed as follows:

$$\text{CA}(F_{Text}, Fuse) = \text{Softmax}\left(\frac{Q_{Text}K_{Fuse}}{\sqrt{d_k}}\right)V_{Fuse} \quad (4)$$

where $\text{CA}(\cdot)$ and $\sqrt{d_k}$ are the cross-attention function and the dimension of the key, respectively, F_{Text} denotes the text feature.

C. Local Text Perception Module

To capture finer details in degraded scenes and more accurately restore original pixel information, we introduce LTPM and integrate it into the feature extraction process of the network. Figure 1 illustrates the detailed workflow of LTPM. The LTPM module first takes input from the fused features and processes it through a two-layer SEBlock, enhancing local feature representation by emphasizing important features and reducing redundant information via an adaptive channel attention mechanism. Drawing on the Text-IF [2], we incorporate the image features generated by the CLIP encoder with the current fused features through scaling and bias control. Specifically, the CLIP-encoded image features are projected into the semantic space of the fused features through an MLP layer, effectively embedding the global image information into the fusion process. Assuming the scaling and bias control parameters obtained from the MLP are denoted as γ and β , the process can be represented as follows:

$$\hat{Fuse} = (1 + \gamma) \odot X + \beta \quad (5)$$

where $\odot$ represents the element-by-element multiplication operation and X represents the fusion features obtained by RSSB. The processed features are then input into a cross-attention module, accompanied by detailed textual features encoded via BLIP, to enhance the integration of multi-modality information from both visual and textual sources. This cross-attention mechanism maps image and text data into a unified feature space, effectively capturing their deep relationships. Given the potential redundancy from the added text details, the module employs dilated convolution to extract multi-scale features at three different dilation rates. Finally, these three layers of multi-scale features are concatenated to form the output of the module.

D. Loss Functions

Due to the strong semantic perception capabilities of VLMs, some prior studies [56], [57] have leveraged them to guide model training. For example, Xu et al. [56] employed CLIP to introduce a weather-prompt learning loss, while Chen et al. [57] proposed an image-perception prompt learning strategy. However, these approaches [56], [57] are limited to image restoration and enhancement tasks, lacking generalization to IVIF, and there is a lack of clean semantic text descriptions corresponding to multi-modality images, particularly for those captured under adverse weather conditions.

To enhance the semantic features of fused images, we propose a novel VLM-driven loss. As shown in Figure 1, this method inputs the fusion result into the CLIP image encoder, while the description of the clean multimodality images is processed by the text encoder, aligning visual and textual features within a shared feature space. By calculating the cosine similarity between these two feature sets, the method effectively measures content consistency between the image and text. The VLM-driven loss $\mathcal{L}_{VLM}$ is expressed as follows:

$$\mathcal{L}_{VLM} = 1 - \frac{F_{Img} \cdot F_{Text}^{Clean}}{\|F_{Img}\| \|F_{Text}^{Clean}\|} \quad (6)$$

where F_{Img} and F_{Text}^{Clean} denote the features obtained from fusion results and text for clean multi-modality images after CLIP encoding respectively, and $\|\cdot\|$ represents the norm.

In addition, we introduce pixel-level losses to ensure that the network generates high-fidelity fusion results. These include color consistency loss $\mathcal{L}_{Color}$, L1 loss $\mathcal{L}_{L1}$, and structural similarity measure loss $\mathcal{L}_{SSIM}$. Letting Img_F represent the fusion result and Img_m^{Clean} represent the clean multi-modality images, the loss $\mathcal{L}_{Color}$ can be expressed as follows:

$$\mathcal{L}_{Color} = \frac{1}{HW} \|T_{CbCr}(Img_F) - T_{CbCr}(Img_m^{Clean})\|_1 \quad (7)$$

where T_{CbCr} denotes the transfer function of RGB to CbCr. $\mathcal{L}_{L1}$ can be expressed as follows:

$$\mathcal{L}_{L1} = \frac{1}{HW} \sum_{m \in \{ir, vi\}} \|Img_F - Img_m^{Clean}\|_1 \quad (8)$$

$\mathcal{L}_{SSIM}$ can be expressed as follows:

$$\mathcal{L}_{SSIM} = 2 - \sum_{m \in \{vi, ir\}} \text{SSIM}(Img_F, Img_m^{Clean}) \quad (9)$$

where $\text{SSIM}(\cdot)$ denotes the structural similarity measurement operator. The total loss $\mathcal{L}_{Total}$ is calculated as follows:

$$\mathcal{L}_{Total} = \mathcal{L}_{VLM} + \mathcal{L}_{Color} + \mathcal{L}_{L1} + \mathcal{L}_{SSIM} \quad (10)$$

E. AWMM-Text Benchmark Construction and Quality Control

We construct a large-scale text-annotated benchmark, AWMM-Text, to support MMIF research under adverse weather. AWMM-Text provides paired textual annotations for each sample in two settings:

- **Degraded setting:** global captions and local detailed descriptions for degraded multi-modality image pairs.
- **Clean setting:** global captions and local detailed descriptions for the corresponding clean image pairs.

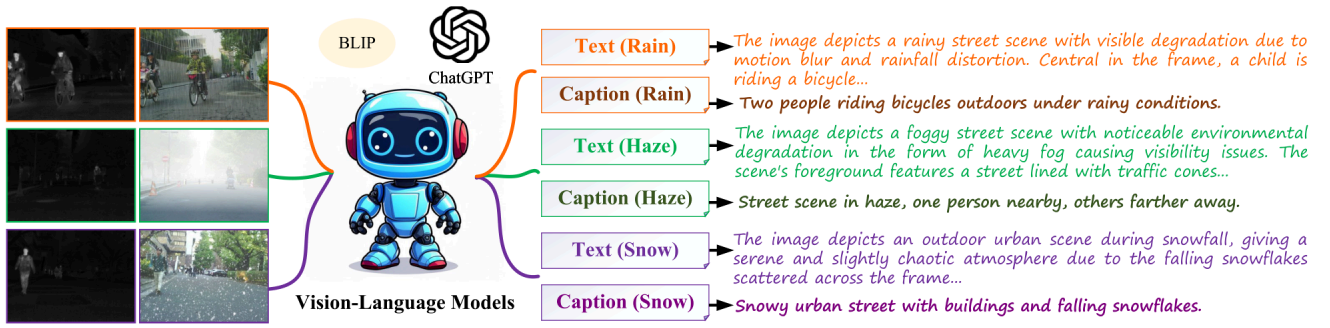


Fig. 2. Text descriptions of different multi-modality images in adverse weather conditions.

To ensure reliable textual supervision for our fusion framework, we standardize the generation and selection of both global captions and detailed descriptions. Detailed descriptions are generated by ChatGPT-4 and truncated to meet the maximum token length of the CLIP text encoder, while global captions are generated by BLIP. The quality of all generated texts is primarily verified through manual screening. In total, we generated detailed and global textual descriptions for 8,500 multi-modality images captured under adverse weather conditions, along with their corresponding clean images.

For quality assurance, we adopt a stratified random sampling strategy and select approximately 30% of the generated texts for manual inspection. Each sampled text is judged with clear pass/fail rules. We reject a text if it (i) describes content that is not observable from the paired infrared and visible images (e.g., guessing identities or unseen facts), (ii) misdescribes key objects or the adverse-weather type, or places them in wrong regions (left/right/near/far), (iii) misses clearly visible important targets such as pedestrians, vehicles, traffic signs, or guardrails, (iv) breaks the required format/length (e.g., not a single paragraph, or truncation makes the description incomplete), or (v) is too generic and provides little useful guidance (e.g., ignores what is clearer in infrared vs. visible when the difference is evident). Furthermore, we use the same checklist across all samples and prioritize correcting salient objects and degradation descriptions. For rejected cases, we follow a simple correction pipeline: we first try regeneration by re-prompting the model with stricter constraints (e.g., explicitly reminding “no guessing” and “describe left/right/near/far”), and if the regenerated text still fails, we perform minimal manual edits to remove hallucinated phrases and fix obvious mismatches. Samples that remain problematic after these steps are discarded to avoid introducing noisy supervision. Figure 2 illustrates textual descriptions corresponding to multi-modality images captured under severe weather conditions.

IV. EXPERIMENT

A. Experimental Setups

1) *Dataset and Comparison Methods*: We evaluate AWM-Fuse on the AWMM-100k dataset¹ [8], and randomly select 50 image pairs for each adverse weather type (rain, haze, and snow). AWMM-100k includes both real-world and synthetic adverse-weather samples by integrating multiple public

RGB/IR datasets with newly collected real data. Notably, the synthetic samples are generated from real paired images using physically plausible degradation processes, so the resulting weather effects closely resemble real observations. Therefore, evaluation on AWMM-100k provides a reliable testbed and reflects the ability of the method to generalize to practical adverse-weather conditions.

We compare the fusion results qualitatively and quantitatively with nine state-of-the-art methods, including GIFNet [58], FreeFusion [59], TIM [33], DRMF [25], FILM [10], EMMA [3], SHIP [24], Text-IF [2] and AWFusion [8]. These baselines cover three representative categories: standard IVIF models (GIFNet, FreeFusion, TIM, EMMA, and SHIP), recent text-guided fusion methods (FILM and Text-IF), and methods targeting complex or adverse scenarios (DRMF, Text-IF and AWFusion), thereby providing a comprehensive comparison under weather degradations. GIFNet, FreeFusion, TIM, and DRMF use clean visible and infrared images as input, while FILM, EMMA, SHIP, and Text-IF rely on visible images restored by two baselines (DehazeFormer [60] and DA-CLIP [12]) and clean infrared images. Although DRMF and Text-IF are designed for complex scenarios, they are not applicable to adverse weather, so we evaluated them in a comparative manner consistent with other normal fusion methods. In contrast, AWFusion and AWM-Fuse process degraded visible and infrared images directly.

For ideal scenes, we conduct experiments on three standard datasets (M3FD [23], MSRS [61] and LLVIP [62]). Since these datasets do not contain degradation, all methods are directly based on the original multi-modal images for fusion comparisons.

2) *Objective Metrics*: We evaluate fusion quality using seven objective metrics. Q_G (Gradient-Based Fusion Performance) [63] measures how well gradient and edge details from the source images are preserved in the fused image. Q_M (Image Fusion Metric Based on a Multiscale Scheme) [63] evaluates fusion quality across multiple scales, capturing both coarse structures and fine textures. Q_S (Piella’s Metric) [63] assesses local structure preservation by comparing the fused image with the source images within local windows. Q_{CV} (Chen–Varshney Metric) [63] emphasizes region-level perceptual consistency by accounting for saliency and local contrast. SCD (Sum of the Correlation of Differences) [64] quantifies complementary information transfer by correlating

¹<https://ixilai.github.io/AWMM-100K/>

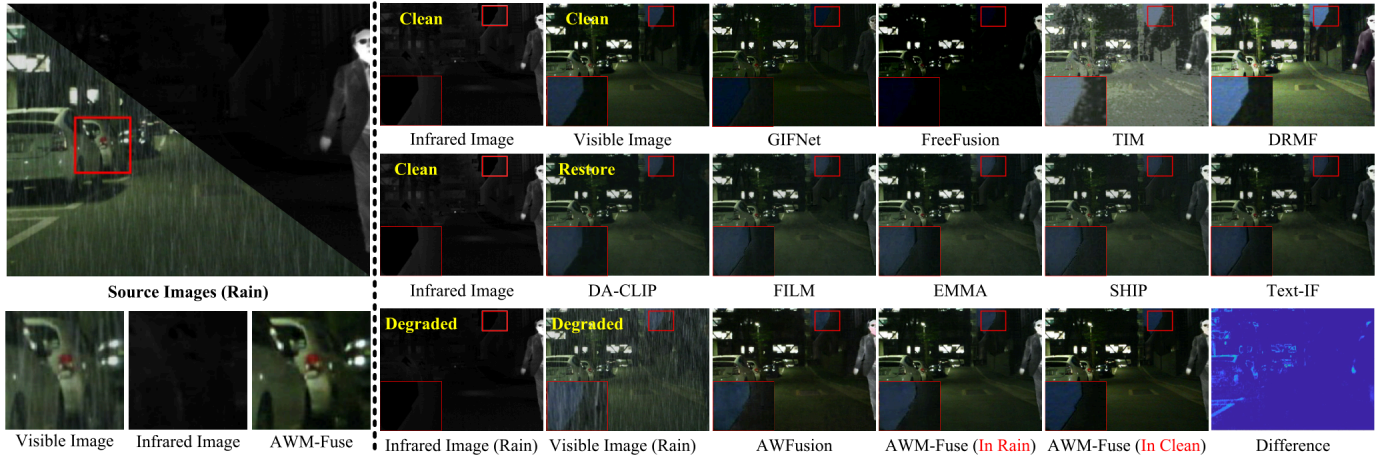


Fig. 3. Comparison of image fusion results of different methods in ideal and rain scenes. The “Difference” represents the difference map between AWM-Fuse (In clean) and AWM-Fuse (In Rain).

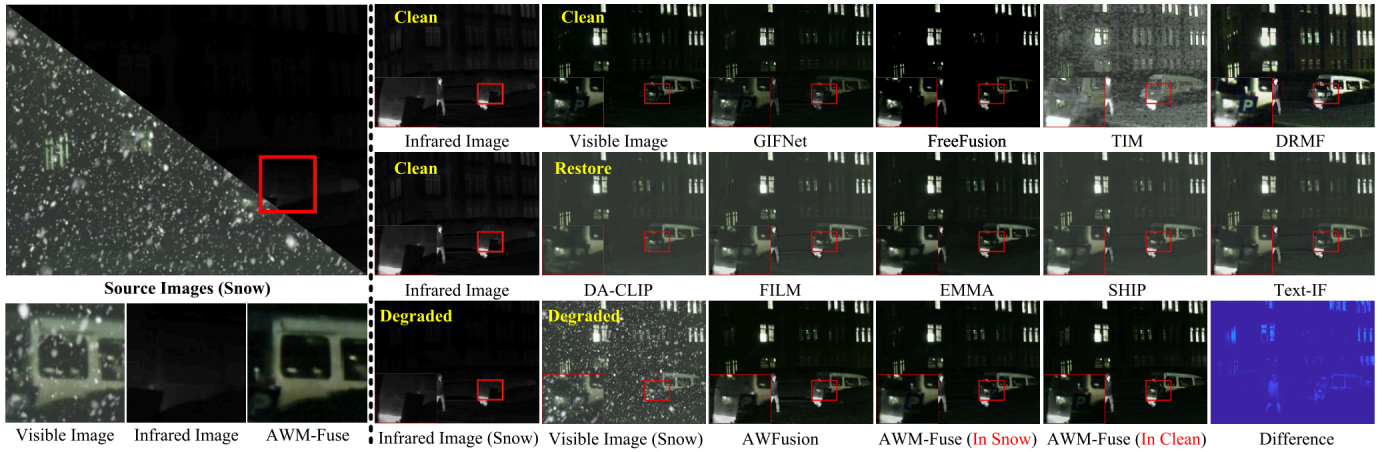


Fig. 4. Comparison of image fusion results of different methods in ideal and snow scenes. The “Difference” represents the difference map between AWM-Fuse (In clean) and AWM-Fuse (In Snow).

difference images between the fused image and the source images. *VIF* (Visual Information Fidelity) [65] measures perceptual quality from an information-fidelity perspective. It estimates how much visual information from a reference is preserved in the fused image under a human-vision-inspired model. *SSIM* (Structural Similarity Index Measure) measures structural similarity between the fused image and reference images in terms of luminance, contrast, and structure. Except for Q_{CV} , higher values indicate better fusion quality. Jointly reporting gradient/multi-scale detail (Q_G , Q_M), structural consistency (Q_S , *SSIM*), perceptual fidelity (Q_{CV} , *VIF*), and information transfer (*SCD*) provides a reliable assessment of fusion quality under complex weather conditions.

3) *Training Setups*: During training, we extracted 700 images for each type of adverse weather (rain, haze, and snow) from the AWM-100k dataset, combining them into a single dataset of 2100 image pairs. The training samples are mixed in a fixed cyclic order, i.e., the sequence is arranged as *rain* $\rightarrow$ *haze* $\rightarrow$ *snow* repeatedly, so that every three consecutive samples correspond to rain, haze, and snow, respectively. Each image was randomly cropped to 160×160 pixels, with a batch size of 2. The Adam optimizer was used with an initial

learning rate of 1×10^{-3} , and the model was trained for over 300 epochs. For network configuration, our RSSBs employ a U-Net structure, with the block configuration of each layer set to [8, 10, 10, 12, 10, 10, 8], achieving a balance between performance and parameter efficiency. The number of feature channels in the RSSBs progressively increase from 48 to 384 and then decrease back to 48.

B. Image Fusion in Adverse Weather

1) *Qualitative Comparison*: As shown in Figure 3 to Figure 5, we compare the proposed algorithm with all comparison methods under three adverse weather scenes. The left part of each figure primarily highlights the interference removal performance of the proposed method. In the right part of each figure, the first row displays fusion results of the comparison methods in ideal conditions, the second row shows fusion with the image restoration method, and the third row presents fusion results from an integrated model designed for adverse weather. We also evaluate our algorithm on clean source images, referred to as AWM-Fuse (In clean), to demonstrate its performance under ideal conditions. Difference maps

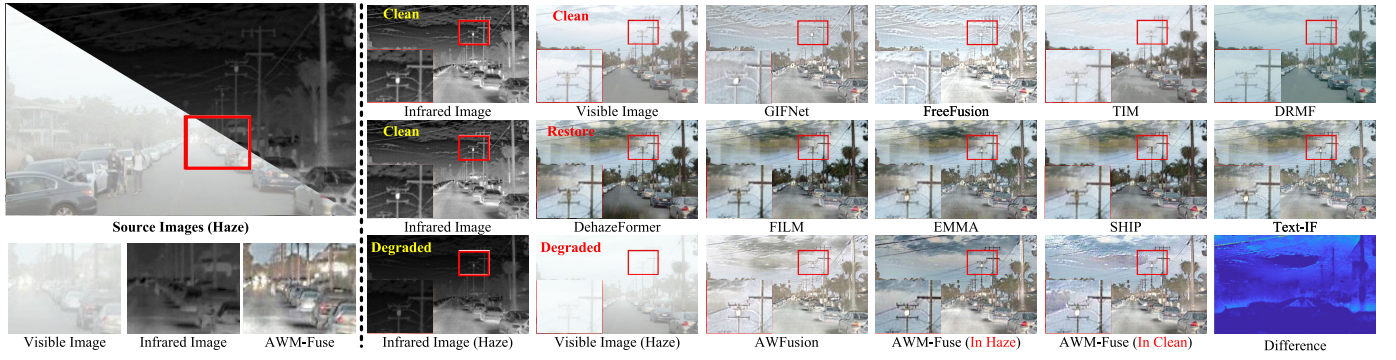


Fig. 5. Comparison of image fusion results of different methods in ideal and haze scenes. The “Difference” represents the difference map between AWM-Fuse (In clean) and AWM-Fuse (In Haze).

TABLE I
COMPARISON OF QUANTITATIVE RESULTS OF DIFFERENT METHODS IN IDEAL AND ADVERSE WEATHER SCENES.
THE BEST SCORES ARE IN RED, WHILE THE SECOND-BEST SCORES ARE IN BLUE

Source	Methods	Pub.	Restoration	$Q_G \uparrow$	$Q_M \uparrow$	$Q_S \uparrow$	$Q_{CV} \downarrow$	$VIF \uparrow$	$SSIM \uparrow$	$SCD \uparrow$
Clean	GIFNet [58]	CVPR 2025	×	0.3555	0.4044	0.8240	529.5149	0.3329	0.4388	1.3872
	FreeFusion [59]	TPAMI 2025		0.2290	0.4207	0.3941	480.6730	0.2627	0.1542	1.4186
	TIM [33]	TPAMI 2024		0.2057	0.3498	0.4630	827.1437	0.1221	0.1585	0.9651
	DRMF [25]	MM 2024		0.3108	0.4170	0.5844	929.2306	0.2579	0.2743	0.8937
	FILM [10]	ICML 2024		0.3522	0.5097	0.7963	560.7920	0.2512	0.3047	1.3869
Rain	EMMA [3]	CVPR 2024	DA-CLIP [12]	0.2812	0.4131	0.8019	493.5733	0.2418	0.3013	1.4011
	SHIP [24]	CVPR 2024		0.3139	0.4105	0.7575	584.6434	0.2354	0.2805	1.2948
	Text-IF [2]	CVPR 2024		0.3771	0.4645	0.8231	549.9020	0.2784	0.3591	1.4088
	AWFusion [8]	INFFus 2026		0.3334	0.6428	0.8335	476.3436	0.2997	0.3528	1.6010
	AWM-Fuse	-	×	0.4321	0.7209	0.8538	452.2425	0.3574	0.4156	1.6527
	GIFNet [58]	CVPR 2025	×	0.3974	0.3651	0.7644	367.2503	0.2781	0.4191	1.4172
	FreeFusion [59]	TPAMI 2025		0.4075	0.4500	0.5463	289.9120	0.2765	0.2842	1.6698
Haze	TIM [33]	TPAMI 2024		0.3020	0.3983	0.6499	507.1376	0.1834	0.2833	0.8978
	DRMF [25]	MM 2024		0.5203	0.6138	0.7530	367.8078	0.3586	0.4182	1.0414
	FILM [10]	ICML 2024	DehazeFormer [60]	0.4452	0.4892	0.7923	395.0446	0.2803	0.3880	1.1668
	EMMA [3]	CVPR 2024		0.3993	0.4341	0.7782	386.5206	0.2749	0.3869	1.1894
	SHIP [24]	CVPR 2024		0.4200	0.4347	0.7753	388.9155	0.2697	0.3736	1.1580
	Text-IF [2]	CVPR 2024		0.4422	0.4706	0.7979	381.8256	0.2895	0.4062	1.2206
	AWFusion [8]	INFFus 2026	×	0.4283	0.5471	0.7738	462.8585	0.3433	0.3901	1.1719
	AWM-Fuse	-		0.5297	0.8193	0.8392	168.0220	0.3936	0.4631	1.3416
Source	Methods	Pub.	Restoration	$Q_G \uparrow$	$Q_M \uparrow$	$Q_S \uparrow$	$Q_{CV} \downarrow$	$VIF \uparrow$	$SSIM \uparrow$	$SCD \uparrow$
Clean	GIFNet [58]	CVPR 2025	×	0.3671	0.3668	0.7846	471.4843	0.2898	0.4269	1.3956
	FreeFusion [59]	TPAMI 2025		0.2965	0.4367	0.4682	356.6543	0.2531	0.2173	1.6087
	TIM [33]	TPAMI 2024		0.2433	0.3626	0.5181	820.5891	0.1505	0.2105	0.8977
	DRMF [25]	MM 2024		0.3948	0.4684	0.6747	729.0171	0.2827	0.3417	1.2065
	FILM [10]	ICML 2024		0.2959	0.4299	0.6296	504.9559	0.2189	0.2491	1.2575
Snow	EMMA [3]	CVPR 2024	DA-CLIP [12]	0.2490	0.3644	0.7310	462.8019	0.1934	0.2472	1.3021
	SHIP [24]	CVPR 2024		0.3400	0.3601	0.5625	505.2532	0.2158	0.2417	1.1179
	Text-IF [2]	CVPR 2024		0.3508	0.4162	0.6444	448.5905	0.2549	0.2906	1.3108
	AWFusion [8]	INFFus 2026		0.3552	0.5535	0.8119	270.1595	0.3024	0.3734	1.4937
	AWM-Fuse	-	×	0.4499	0.7046	0.8456	217.7734	0.3573	0.4406	1.6263
	GIFNet [58]	CVPR 2025	×	0.3671	0.3668	0.7846	471.4843	0.2898	0.4269	1.3956
	FreeFusion [59]	TPAMI 2025		0.2965	0.4367	0.4682	356.6543	0.2531	0.2173	1.6087
	TIM [33]	TPAMI 2024		0.2433	0.3626	0.5181	820.5891	0.1505	0.2105	0.8977
	DRMF [25]	MM 2024		0.3948	0.4684	0.6747	729.0171	0.2827	0.3417	1.2065

are provided to illustrate variations between fusion results obtained with clean images and under adverse weather.

The comparison indicates that, under ideal conditions, all fusion methods capture features from different modalities. Although preprocessing degraded visible images can reduce interference from degraded data, this approach often relies heavily on the initial restoration, as shown in Figure 3 to Figure 5, where unresolved degradation affects fusion. AWFusion mitigates adverse weather effects, but residual artifacts from color distortion and incomplete degradation removal persist.

In contrast, AWM-Fuse excels in degradation removal, color fidelity, and multi-modality information extraction. The difference maps confirm that our algorithm preserves most salient scene features. Finally, fusion results in clean images show that, despite training under adverse weather, our algorithm effectively learns multi-modality feature extraction, maintaining high-quality fusion in ideal conditions.

2) *Quantitative Comparison:* Table I presents the quantitative comparison results of different methods under three types of adverse weather degradation. Our method ranks

TABLE II
QUANTITATIVE COMPARISON OF ALL METHODS IN THREE STANDARD DATASETS. THE BEST SCORES ARE IN RED,
WHILE THE SECOND-BEST SCORES ARE IN BLUE

Datasets	Methods	Pub.	$Q_G \uparrow$	$Q_M \uparrow$	$Q_S \uparrow$	$Q_{CV} \downarrow$	$VIF \uparrow$	$SSIM \uparrow$	$SCD \uparrow$
M3FD	GIFNet [58]	CVPR 2025	0.4446	0.4538	0.8270	527.2287	0.2972	0.4598	1.9405
	FreeFusion [59]	TPAMI 2025	0.3818	0.4071	0.8535	533.7730	0.2838	0.4505	1.6988
	TIM [33]	TPAMI 2024	0.3447	0.4325	0.7928	1135.7788	0.2678	0.3553	0.4246
	DRMF [25]	MM 2024	0.343	0.4707	0.8041	975.7381	0.3808	0.3493	0.8389
	FILM [10]	ICML 2024	0.4892	1.0778	0.8559	526.8890	0.3840	0.4568	1.2927
	EMMA [3]	CVPR 2024	0.4665	0.5598	0.8501	529.0906	0.3839	0.4644	1.3918
	SHIP [24]	CVPR 2024	0.5572	1.5184	0.8492	519.7535	0.3897	0.4476	1.1945
	Text-IF [2]	CVPR 2024	0.5344	1.0641	0.8724	448.7277	0.4173	0.4901	1.4025
	AWFusion [8]	INFFus 2026	0.4538	0.5680	0.8420	544.1436	0.3723	0.4657	1.1963
	AWM-Fuse	-	0.5752	1.5261	0.8619	503.9105	0.4439	0.5036	1.2467
Datasets	Methods	Pub.	$Q_G \uparrow$	$Q_M \uparrow$	$Q_S \uparrow$	$Q_{CV} \downarrow$	$VIF \uparrow$	$SSIM \uparrow$	$SCD \uparrow$
MSRS	GIFNet [58]	CVPR 2025	0.3974	0.3651	0.7644	367.2503	0.2781	0.4191	1.4172
	FreeFusion [59]	TPAMI 2025	0.4075	0.4500	0.5463	289.9120	0.2765	0.2842	1.5298
	TIM [33]	TPAMI 2024	0.302	0.3983	0.6499	507.1376	0.1834	0.2833	0.8978
	DRMF [25]	MM 2024	0.5203	0.6138	0.753	367.8687	0.3586	0.4182	1.0414
	FILM [10]	ICML 2024	0.6715	1.6424	0.8521	148.8272	0.4387	0.4791	1.4297
	EMMA [3]	CVPR 2024	0.5851	0.7199	0.8385	140.2975	0.4244	0.4758	1.5062
	SHIP [24]	CVPR 2024	0.6280	1.7853	0.8322	132.7968	0.4166	0.4608	1.3840
	Text-IF [2]	CVPR 2024	0.6465	1.4406	0.8546	130.9396	0.4487	0.4909	1.5675
	AWFusion [8]	INFFus 2026	0.5540	0.8039	0.8466	178.0195	0.4187	0.4775	1.2941
	AWM-Fuse	-	0.6531	1.9846	0.8512	127.9056	0.4641	0.5042	1.5365
Datasets	Methods	Pub.	$Q_G \uparrow$	$Q_M \uparrow$	$Q_S \uparrow$	$Q_{CV} \downarrow$	$VIF \uparrow$	$SSIM \uparrow$	$SCD \uparrow$
LLVIP	GIFNet [58]	CVPR 2025	0.3916	0.1829	0.7641	656.5176	0.2284	0.4249	1.5145
	FreeFusion [59]	TPAMI 2025	0.3241	0.1533	0.5703	579.8194	0.1964	0.3028	1.6986
	TIM [33]	TPAMI 2024	0.3149	0.1637	0.6415	764.2835	0.2337	0.2906	0.7223
	DRMF [25]	MM 2024	0.3579	0.1576	0.6095	842.8371	0.2363	0.3166	1.1763
	FILM [10]	ICML 2024	0.6796	1.0252	0.8860	395.3111	0.3905	0.4842	1.5251
	EMMA [3]	CVPR 2024	0.5630	0.3088	0.8639	406.5657	0.3452	0.4822	1.5836
	SHIP [24]	CVPR 2024	0.6628	1.3961	0.8741	388.4069	0.3792	0.4786	1.5028
	Text-IF [2]	CVPR 2024	0.6479	0.8679	0.8752	377.1400	0.4046	0.4942	1.5458
	AWFusion [8]	INFFus 2026	0.5843	0.4942	0.8701	426.4772	0.3658	0.4854	1.5968
	AWM-Fuse	-	0.6806	1.1895	0.8937	382.5485	0.4014	0.5135	1.5889

TABLE III
COMPARING QUANTITATIVE RESULTS OF DIFFERENT METHODS IN SEMANTIC SEGMENTATION TASK.
THE BEST SCORES ARE IN RED, WHILE THE SECOND-BEST SCORES ARE IN BLUE

Source	Methods	Pub.	Restoration	Background	Car	Person	Bike	Curve	Cuadrail	Color cone	Bump	mIoU
Clean	GIFNet [58]	CVPR 2025	×	98.4	88.98	71.13	71.92	59.19	78.05	65.3	64.62	74.69
	FreeFusion [59]	TPAMI 2025		97.93	85.16	71.4	63.16	40.24	78.7	56.57	60.97	69.26
	TIM [33]	TPAMI 2024		97.68	82.22	65.85	58.6	38.75	75.76	53.92	51.65	65.55
	DRMF [25]	MM 2024		98.31	88.11	70.5	70.54	51.75	82.46	63.97	74.93	75.07
Haze	FILM [10]	ICML 2024	DehazeFormer [60]	97.83	85.7	69.11	65.13	52.33	76.11	32.81	40.17	64.89
	EMMA [3]	CVPR 2024		97.76	85.35	66.27	64.98	53.31	75.04	31.98	37.22	63.99
	SHIP [24]	CVPR 2024		98.42	88.57	72.18	69.91	63.14	79.83	62.79	72.6	75.93
	Text-IF [2]	CVPR 2024		98.51	89.42	73.52	70.69	63.52	79.62	63.76	75.91	76.87
	AWFusion [8]	INFFus 2026	×	98.4	89.03	70.91	69.62	60.08	83.15	64.45	73.72	76.17
	AWM-Fuse	-		98.48	89.09	72.41	70.89	63.53	86.38	64.6	75.92	77.66

within the top two across all seven metrics, demonstrating that the proposed algorithm achieves superior performance in addressing multi-modality image fusion challenges under adverse weather conditions. In addition, for the integrated model, we observe that AWFusion performs well in both rain and snow scenes. However, the proposed method outperforms AWFusion in both image quality assessment metric scores and color recovery.

C. Experiments on the Standard Dataset

Although our method is primarily designed for adverse-weather scenarios, we also conduct additional experiments under standard conditions to examine whether the model can

effectively learn cross-modal interaction between infrared and visible images while performing feature restoration. Specifically, we compare fusion performance on three public datasets without weather degradations, including M3FD [23], MSRS [61], and LLVIP [62]. As shown in Table II, our method achieves better scores than most baselines across multiple metrics, indicating that it retains strong fusion capability even in the absence of degradations. Notably, AWFusion tends to focus more on degradation removal, which may weaken cross-modal interaction and lead to inferior performance on standard scenes compared to some conventional fusion methods. In contrast, Text-IF performs well in this setting, largely because text guidance encourages the model to focus more on cross-modal complementarity rather than degradation-related cues. Overall,

TABLE IV
COMPARING QUANTITATIVE RESULTS OF DIFFERENT METHODS IN OBJECT DETECTION TASK. MAP DENOTES MEAN AVERAGE PRECISION. THE BEST SCORES ARE IN RED, WHILE THE SECOND-BEST SCORES ARE IN BLUE

Source	Methods	Pub.	Restoration	People	Car	Bus	Lamp	Motorcycle	Truck	AP@0.5	mAP@[0.5:0.95]
Clean	GIFNet [58]	CVPR 2025	×	0.806	0.906	0.919	0.777	0.671	0.779	0.81	0.506
	FreeFusion [59]	TPAMI 2025		0.813	0.907	0.927	0.787	0.72	0.809	0.828	0.526
	TIM [33]	TPAMI 2024		0.731	0.898	0.878	0.657	0.656	0.769	0.765	0.48
	DRMF [25]	MM 2024		0.699	0.897	0.896	0.787	0.697	0.784	0.793	0.494
Snow	FILM [10]	ICML 2024	DA-CLIP [12]	0.789	0.891	0.888	0.678	0.657	0.743	0.774	0.483
	EMMA [3]	CVPR 2024		0.785	0.894	0.901	0.674	0.68	0.766	0.783	0.484
	SHIP [24]	CVPR 2024		0.794	0.9	0.899	0.754	0.691	0.793	0.805	0.499
	Text-IF [2]	CVPR 2024		0.819	0.907	0.921	0.771	0.723	0.803	0.824	0.522
	AWFusion [8]	INFFus 2026	×	0.791	0.908	0.903	0.781	0.714	0.801	0.816	0.522
	AWM-Fuse	-		0.828	0.913	0.915	0.791	0.717	0.811	0.829	0.527

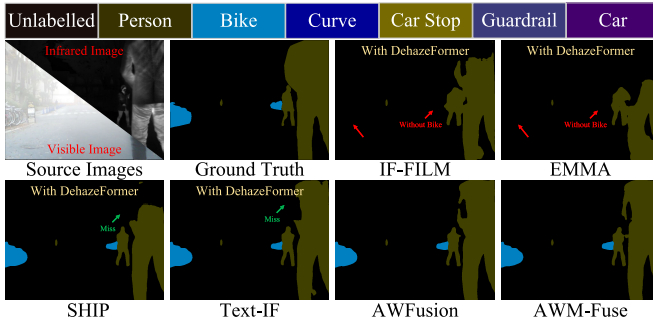


Fig. 6. Segmentation comparisons of different methods under hazy conditions.

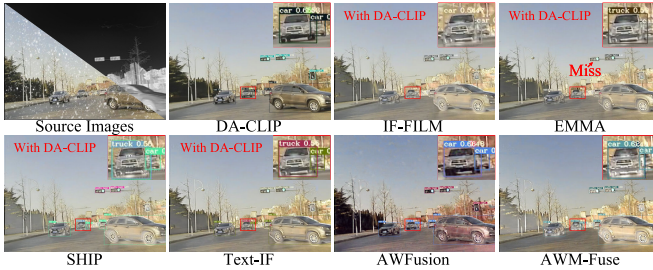


Fig. 7. Detection comparisons of different methods under snow conditions.

these results demonstrate that combining global and local text is not only effective for adverse weather, but also yields stable and strong fusion performance in standard scenarios.

D. Applications in Downstream Tasks

We tested all methods on semantic segmentation and object detection to evaluate fused image quality. We used BANet [66] as the segmentation network and YOLOv7 [6] as the detector. Table III and Table IV present quantitative comparisons of various methods on these tasks. As shown in Table III, our method achieves the highest intersection-over-union (IoU) scores across all categories, as well as the overall mean IoU in semantic segmentation. Furthermore, Table IV shows that the proposed method yields the highest detection accuracy.

In addition, we present the qualitative comparison results. As shown in Figure 6, the semantic information in the source image is diminished due to haze interference. While an image dehazing algorithm (DehazeFormer) was added as a pre-processing step for visible images in the first four comparison

methods, they still produce segmentation errors and missing regions. Figure 7 shows the object detection comparison results. Although the snow removal method (DA-CLIP) was added to some fusion algorithms, the results failed to retain significant details due to the lack of synchronization between restoration and fusion. In contrast, the proposed algorithm effectively preserves scene information and outperforms others in object detection.

E. Ablation Experiment

1) *Effectiveness Analysis of Text*: To further investigate the influence of textual information on network performance, we conduct four ablation experiments:

- **Noisy Text (partial mismatch)**: we introduce *partial* textual noise by instructing the VLM to generate descriptions that contain several incorrect or irrelevant statements (e.g., wrong object attributes or local weather cues), while the remaining content is still consistent with the image.
- **Wrong Scene Text (full mismatch)**: we inject a *fully mismatched* scene description, where the overall scene category and major objects/weather cues are intentionally replaced by an unrelated one, making the text globally inconsistent with the image (i.e., all key content is wrong).
- **Text-Reduced**: we reduce the amount of textual input by approximately half, by truncating the global caption and the detailed description to keep only the most basic scene/weather information, thereby removing many fine-grained semantic cues.
- **Text-Augmented**: we add redundant and verbose sentences by repeating or rephrasing existing descriptions and appending extra minor details, increasing text length without providing new useful information.

From the experimental results in Table V, we observe the following: (1) Training with misguided textual descriptions leads to the most severe degradation in fusion performance; (2) Reducing textual input causes image feature-based metrics (e.g., Q_S and Q_M) to drop significantly, suggesting that fewer textual cues weaken the ability of the network to capture fine details; (3) Adding redundant text improves Q_S and Q_M , but decreases perception-based metrics such as Q_{CV} and VIF , implying that redundancy introduces deviations from natural human visual perception.

Quantitative results under full mismatch: Table VI presents the comparative results under the Wrong Scene

TABLE V
QUANTITATIVE COMPARATIVE STRUCTURES IN ABLATION STUDIES. THE BEST SCORES ARE IN BOLD

Sources	Methods	$Q_G\uparrow$	$Q_M\uparrow$	$Q_S\uparrow$	$Q_{CV}\downarrow$	$VIF\uparrow$	$SSIM\uparrow$	$SCD\uparrow$
Snow	Noise Text	0.4381	0.6053	0.8315	255.5851	0.3322	0.4260	1.5986
	Text-Reduced	0.4312	0.6761	0.8416	231.2314	0.3410	0.4332	1.6054
	Text-Augmented	0.4518	0.7105	0.8432	225.1728	0.3564	0.4345	1.6127
	AWM-Fuse	0.4499	0.7046	0.8456	217.7734	0.3573	0.4406	1.6263
Haze	w/o GTPM	0.3537	0.6714	0.7481	235.3642	0.2812	0.3825	1.1227
	w/o LTPM	0.3298	0.5991	0.6827	227.2447	0.3145	0.3989	1.1892
	w/o Text (ChatGPT)	0.3477	0.6734	0.7395	220.3748	0.3285	0.4137	1.2328
	w/o VLM-driven Loss	0.4477	0.6892	0.6992	192.1521	0.3331	0.4305	1.2358
	AWM-Fuse	0.5297	0.8193	0.8392	168.0220	0.3936	0.4631	1.3416

TABLE VI
QUANTITATIVE ANALYSIS UNDER MISMATCHED WEATHER TEXT PROMPTS. THE BEST SCORES ARE IN BOLD

Source	Methods	$Q_G\uparrow$	$Q_M\uparrow$	$Q_S\uparrow$	$Q_{CV}\downarrow$	$VIF\uparrow$	$SSIM\uparrow$	$SCD\uparrow$	Avg. degradation
Rain	Wrong Scene Text	0.3946	0.6446	0.8449	506.8321	0.3245	0.3921	1.5473	7.66%
	AWM-Fuse	0.4321	0.7209	0.8538	452.2425	0.3574	0.4156	1.6527	
Haze	Wrong Scene Text	0.4432	0.5567	0.7147	440.3820	0.3766	0.3753	1.1262	37.81%
	AWM-Fuse	0.5297	0.8193	0.8392	168.0220	0.3936	0.4631	1.3416	
Snow	Wrong Scene Text	0.4045	0.5918	0.8327	312.2513	0.3165	0.4084	1.4739	14.16%
	AWM-Fuse	0.4499	0.7046	0.8456	217.7734	0.3573	0.4406	1.6263	

Text (full mismatch) setting. The results show that when the textual prompt is completely inconsistent with the image scene, the performance drops most significantly: the average metrics decrease by 7.66%, 37.81%, and 14.16% in rain, haze, and snow scenarios, respectively. Moreover, the mismatched prompts have the greatest impact in haze scenes, followed by snow, while rain shows the smallest degradation. This is mainly because haze degradation is dominated by global atmospheric scattering, which substantially reduces overall contrast and suppresses scene-discriminative information, making the fusion model more reliant on textual semantics to select appropriate enhancement and interaction strategies. In contrast, rain streaks and snow particles are typically local and directionally structured occlusions; even when the prompt is biased, the model can still identify and suppress these degradations, leading to a relatively smaller performance drop.

Qualitative results under full mismatch: To further visually demonstrate the impact of erroneous text on fusion performance, we provide a qualitative comparison in Figure 8(a). It can be observed that when a completely incorrect scene prompt is used, the model finds it more difficult to simultaneously suppress degradations and preserve details and color. For example, in haze scenes, the wrong prompt weakens the ability of the model to recover dense haze in distant regions, leading to noticeable color distortion. Notably, even under fully mismatched prompts, the model does not collapse; it can still preserve key targets from the multi-modality inputs and remove most degradations. These observations indicate that the text mainly guides the model to attend to the degradation type and relevant local regions, while the fused content is still primarily driven by infrared and visible features, rather than being directly determined by the text.

Robustness to temporal prompt bias: Since infrared images often provide limited cues about time of day (e.g., day or night), we further examine the robustness of textual

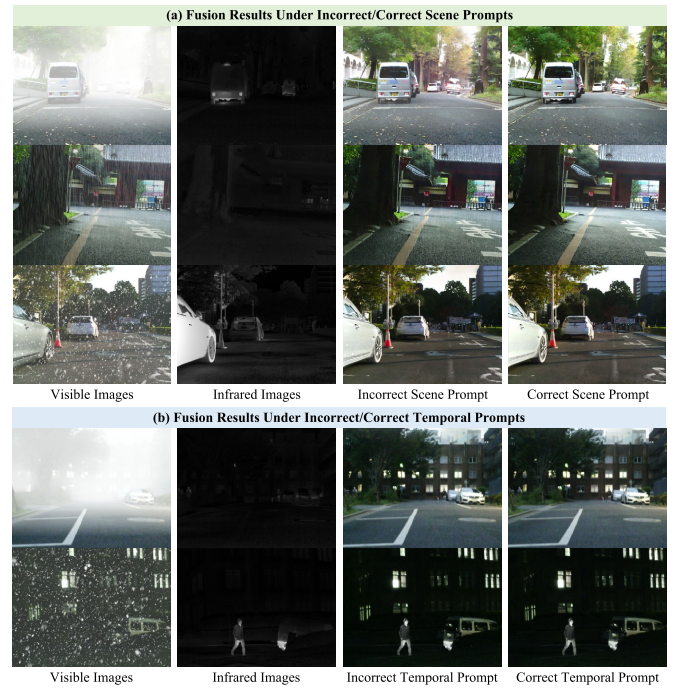


Fig. 8. Qualitative comparison between correct and incorrect text prompts.

guidance in this regard. We artificially inject erroneous temporal cues during fusion (e.g., labeling a nighttime scene as daytime). As shown in Figure 8(b), our method still produces reasonable fusion results despite the biased cues. This demonstrates that text serves primarily as auxiliary guidance, while the main features are extracted from the original image, confirming the robustness.

2) Effectiveness Analysis of GTPM and LTPM: The GTPM module uses global captions (CLIP-encoded) to provide global semantic guidance for degradation identification and overall

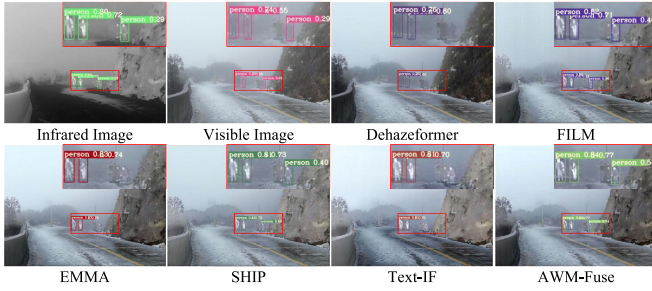


Fig. 9. Fusion results of different algorithms in real-world scenarios.

enhancement strategy. As shown in Table V, removing GTPM (w/o GTPM, i.e., local-only) causes consistent drops, especially on perception/consistency metrics. Under haze, *VIF* decreases from 0.3936 to 0.2812 and *SCD* decreases from 1.3416 to 1.1227, suggesting that without global guidance the model tends to leave more residual degradations and weaker semantic consistency. The LTPM module leverages detailed descriptions to inject local semantic cues, benefiting fine-grained structure and detail recovery in degraded regions. When ablating LTPM (w/o LTPM, i.e., global-only), haze results show larger drops in detail/structure metrics: Q_M drops from 0.8193 to 0.5991 and Q_S drops from 0.8392 to 0.6827, indicating that global captions alone are insufficient for accurate local restoration.

Overall, these results reveal a complementary interaction: GTPM mainly stabilizes global perception and degradation-aware strategy selection (e.g., *VIF*, *SCD*), while LTPM mainly improves local structure/detail recovery (e.g., Q_M , Q_S). Combining both yields the best overall performance, showing the two text branches are mutually supportive rather than redundant.

3) *Effectiveness Analysis of Text (ChatGPT)*: We replaced all the detailed text generated by ChatGPT with caption data as input to the LTPM. As shown in Table V, this replacement results in reduced scores for image feature-based metrics Q_G and Q_M . However, since some text input is still retained, these metrics remain higher than those obtained by completely ablating the LTPM.

4) *Effectiveness Analysis of VLM-Driven Loss*: During training, text data generated by the VLM are used to supervise fusion result generation, ensuring alignment between image and text semantic information. The ablation results in Table V show a degree of performance degradation when this supervision is removed, highlighting the effectiveness of the proposed VLM-driven loss in achieving text-image feature alignment.

F. Extended Experiments in Real-World Scenarios

We conduct fusion experiments on real-haze scenes from the AWM-100k dataset. Since real-world data do not provide degradation-free reference images for quantitative evaluation, we mainly perform qualitative comparisons and additionally employ a YOLOv7 detector to evaluate object preservation in the fused results under severe degradations. As shown in Figure 9, detection on infrared images is clearly better than that on visible images in haze, indicating that the vis-

ible modality suffers from substantial degradation. Moreover, restoring the visible image with DehazeFormer before fusion further reduces detection accuracy, suggesting that the two-stage “restoration + fusion” pipeline may introduce restoration errors into subsequent fusion and even amplify them. Specifically, EMMA and Text-IF miss distant pedestrians in this scenario; FILM and SHIP can detect most objects, but often produce blurred boundaries, which degrades detection accuracy. In contrast, our method better preserves critical targets under real adverse weather and achieves the highest detection accuracy across all object categories.

G. Limitation

Extracting features with VLMs inevitably incurs additional computational overhead. The proposed algorithm, when tested on images of size 480×640 , has FLOPs of 1145.5G and parameters of 137.29M. In comparison, Text-IF, which also incorporates VLM models, has FLOPs of 1518.88G and parameters of 89.01M. We recognize this as a limitation of the current work and plan to optimize computational efficiency in future iterations.

To clarify the overhead introduced by the dual text perception modules, we provide a module-level complexity breakdown under the same input resolution (480×640). Relative to the full model, GTPM introduces about 2.70% additional computation and accounts for about 0.09% of the total parameters. In contrast, LTPM is the major source of overhead, contributing about 19.1% of the full-model computation and about 7.29% of the parameters. Moreover, using the backbone without both modules as the reference, adding GTPM and LTPM results in about 3.45% and 24.44% relative increases in computation, respectively, indicating that LTPM brings a more noticeable computational and parameter burden while enhancing local representation capability.

Overall, despite the non-trivial computational cost, our method still achieves strong fusion performance, outperforming many state-of-the-art image restoration and fusion models in our experiments.

V. CONCLUSION

In this paper, we propose a multi-modality image fusion method tailored for adverse weather. We leverage the image-to-text generation capability of ChatGPT to create text descriptions for source images under adverse weather and use BLIP to generate captions. To utilize this text, we design global and local textual feature perception modules. We also propose a VLM-driven loss to align images and text features within the CLIP feature space. In addition, we build a large-scale text benchmark, AWM-Text, which provides global and local textual annotations for adverse-weather multi-modality image pairs to support text-guided fusion research. Finally, we validate the effectiveness of our method through extensive experiments across adverse weather conditions, clean scenarios, and downstream tasks.

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